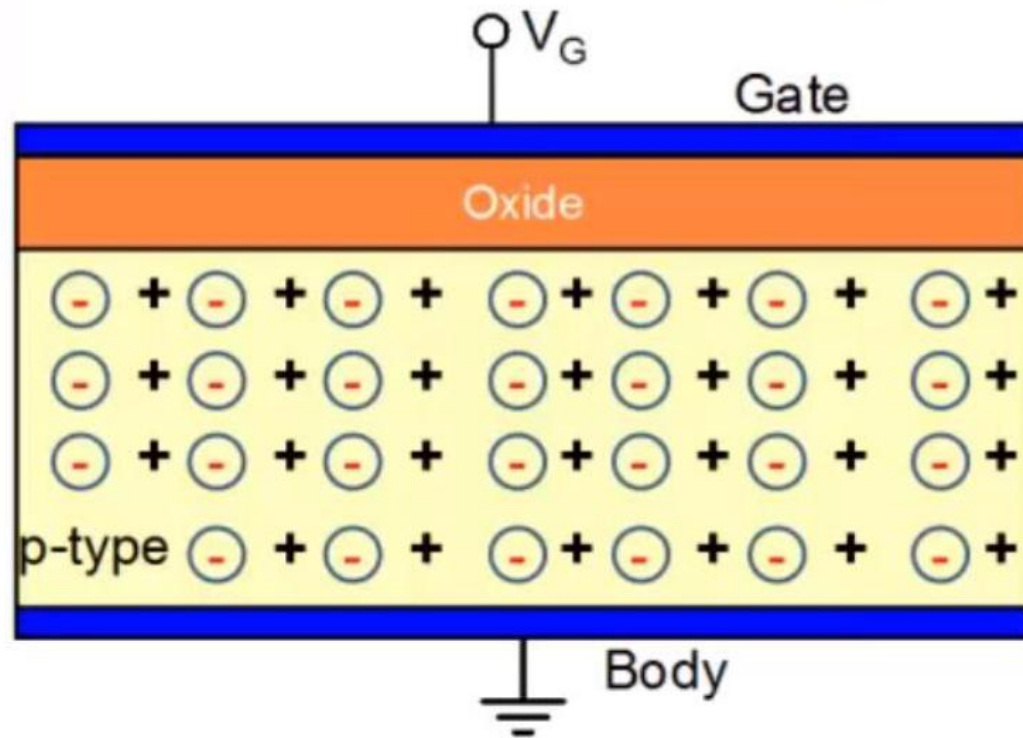


Flat band condition

$$V_G = V_{FB}$$

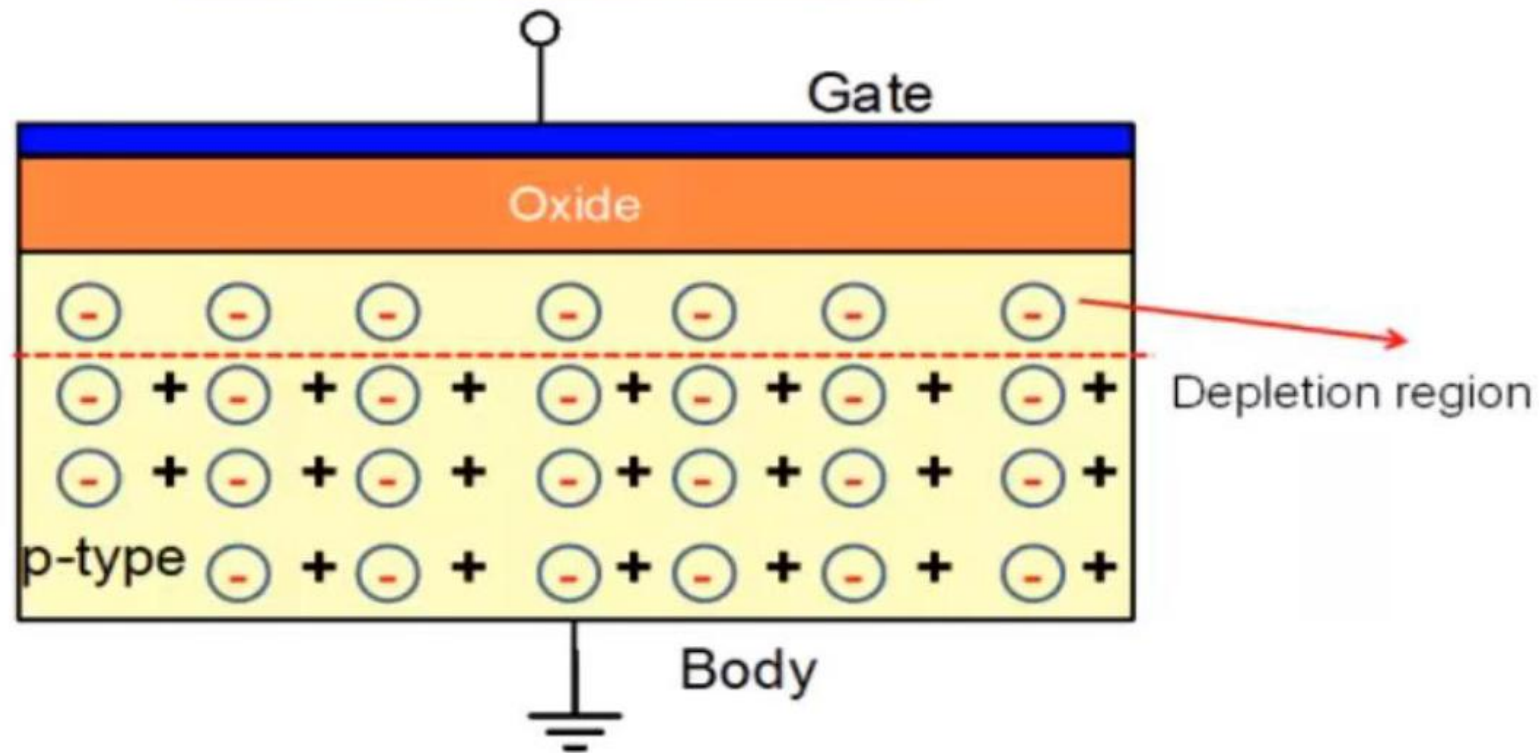


Whenever two different material are brought into contact, an internal potential difference develops like in a pn junction. Thus even when no gate voltage is applied, there is a voltage across the mos capacitor.

$V_G = V_{FB}$; Flat-band condition meaning no NET voltage across the capacitor.
Uniform hole density everywhere

Depletion

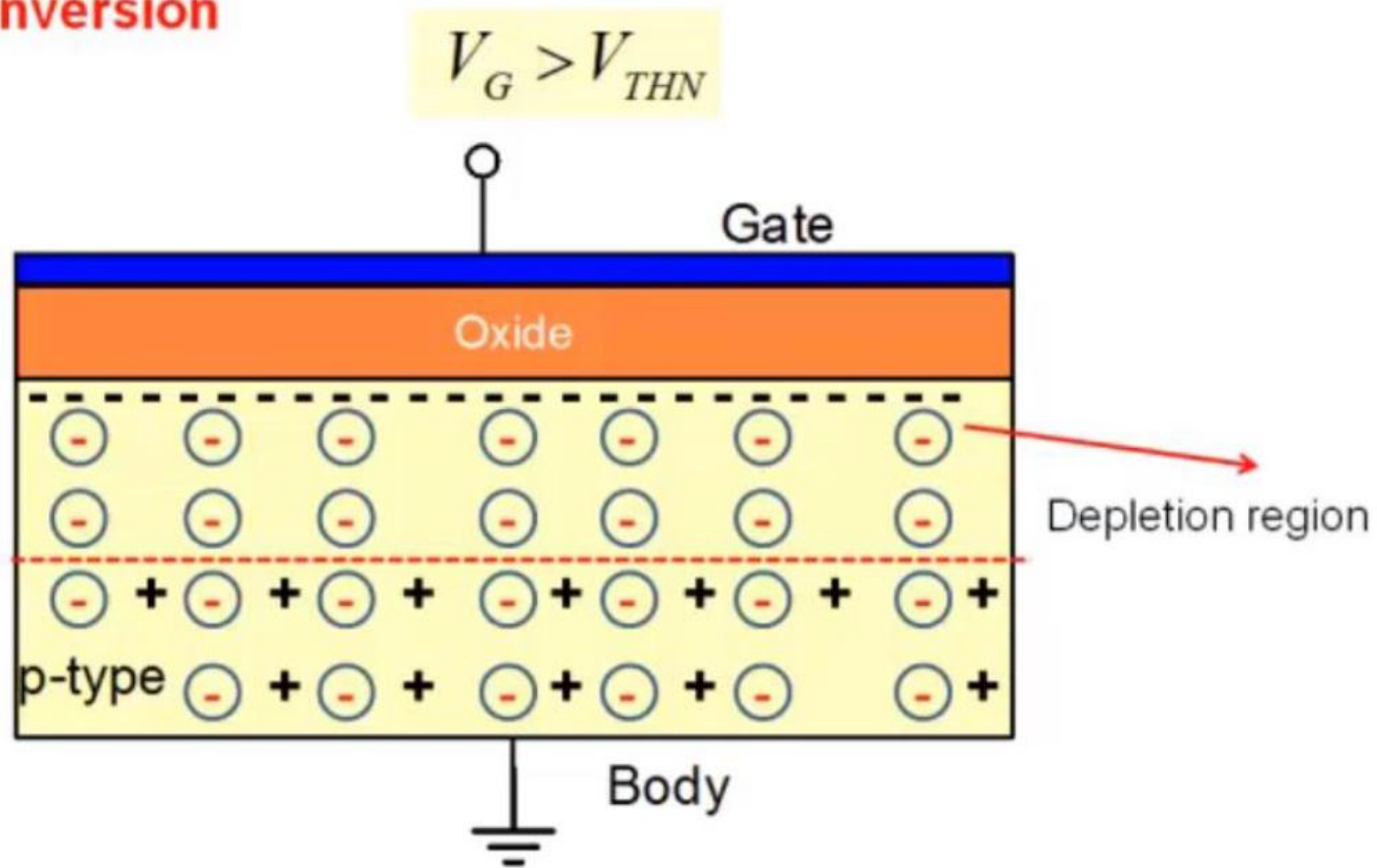
$$V_G > V_{FB} \text{ but } V_G < V_{THN}$$



Holes are depleted from the surface $p_S < p_B$

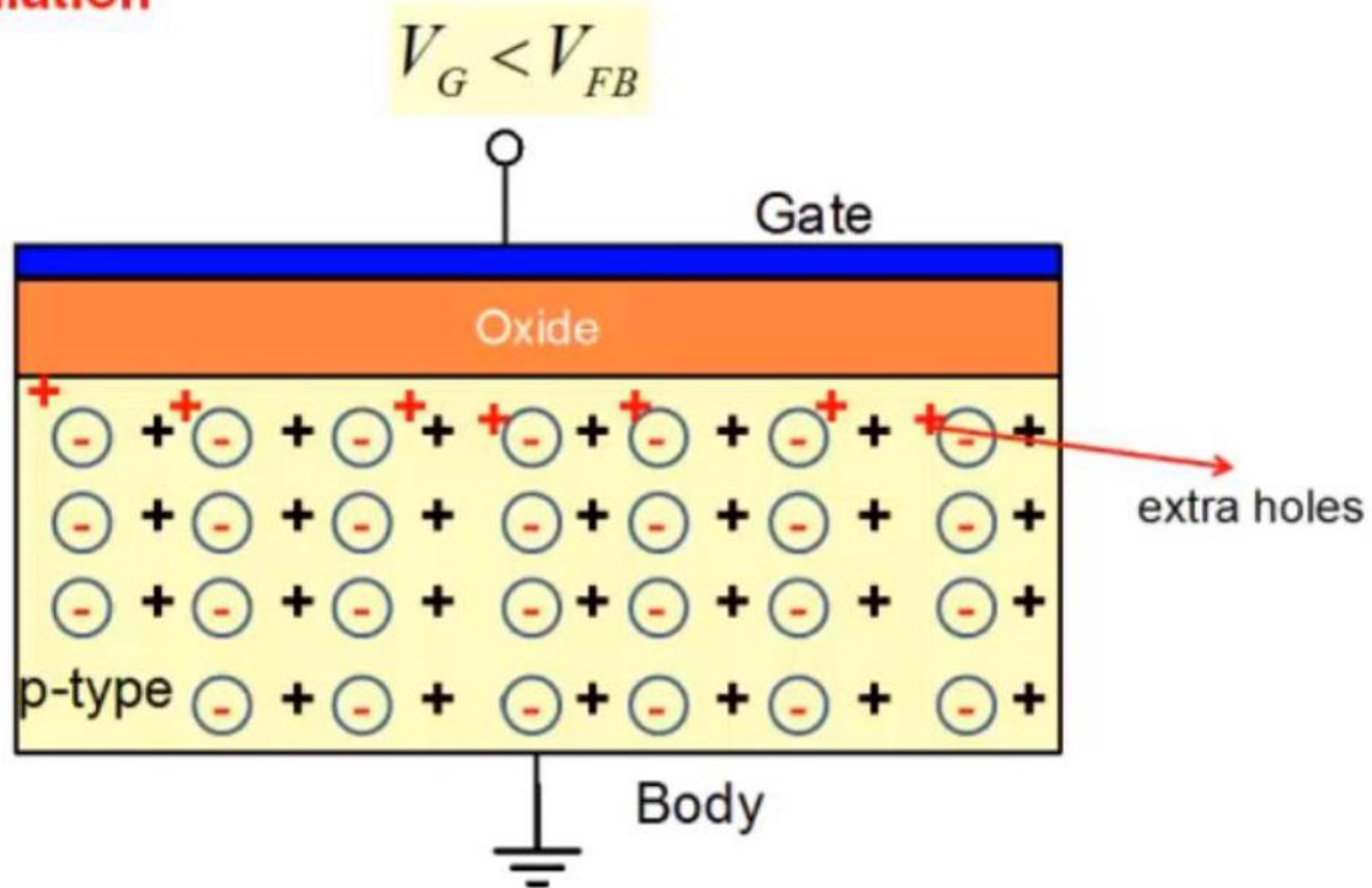
Although $n_S > n_B$ electron density is also very small

Strong Inversion



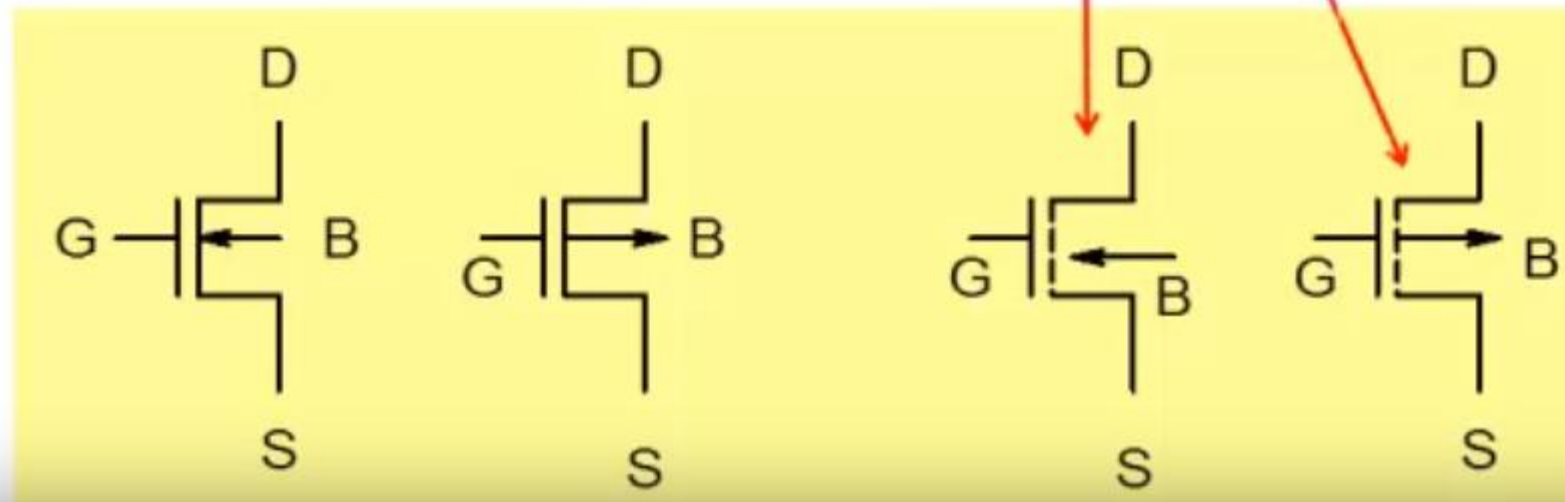
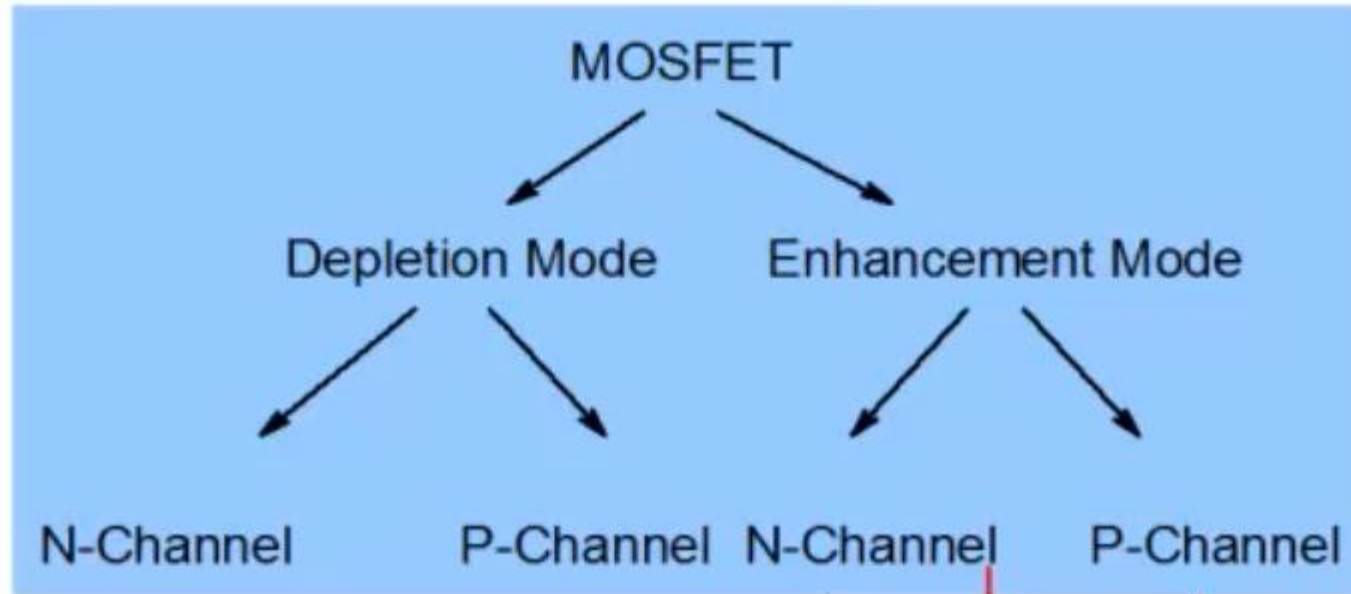
Electrons are accumulated at the surface $n_S \gg N_A$

Accumulation

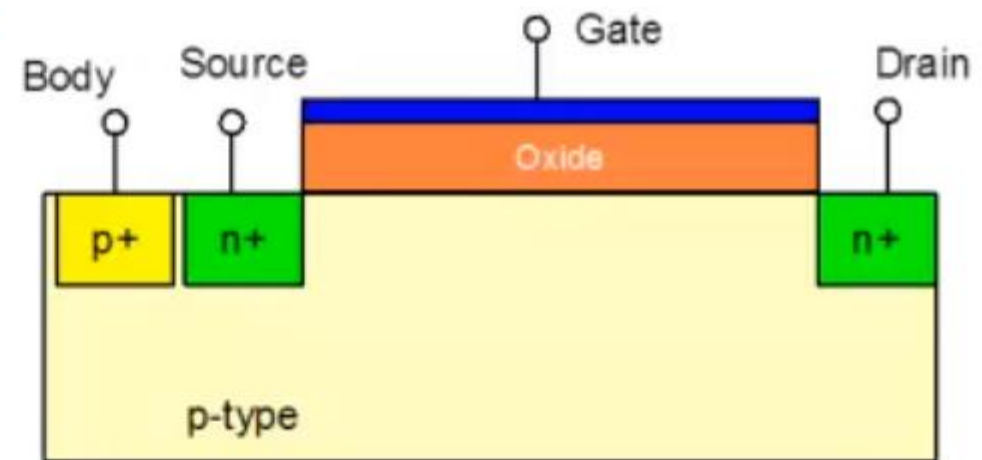
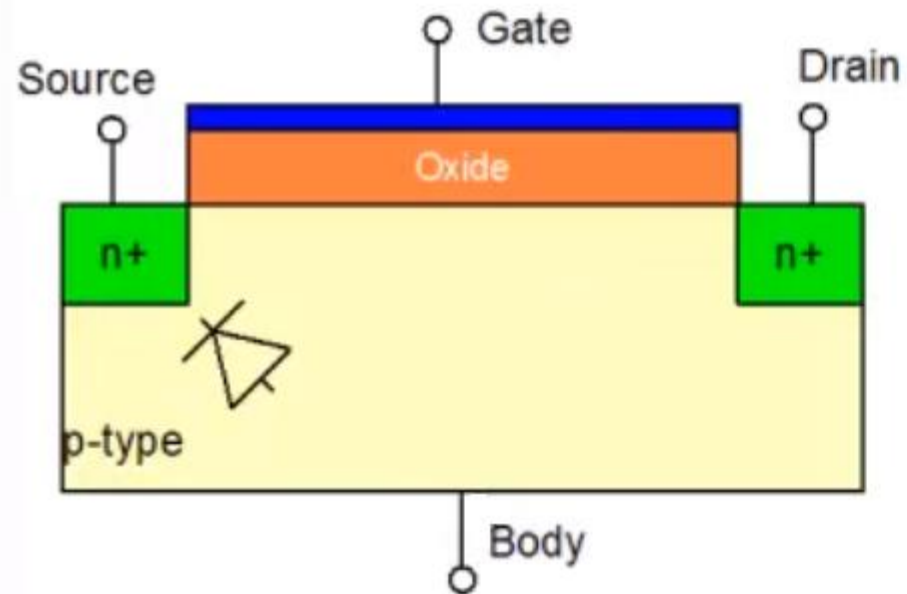
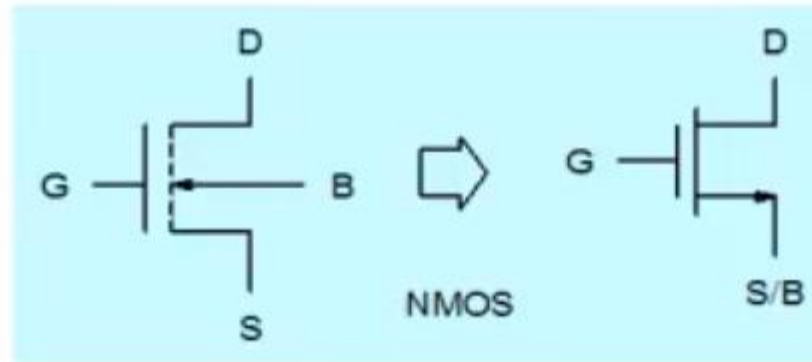


Holes are accumulated at the surface $p_S > p_B$

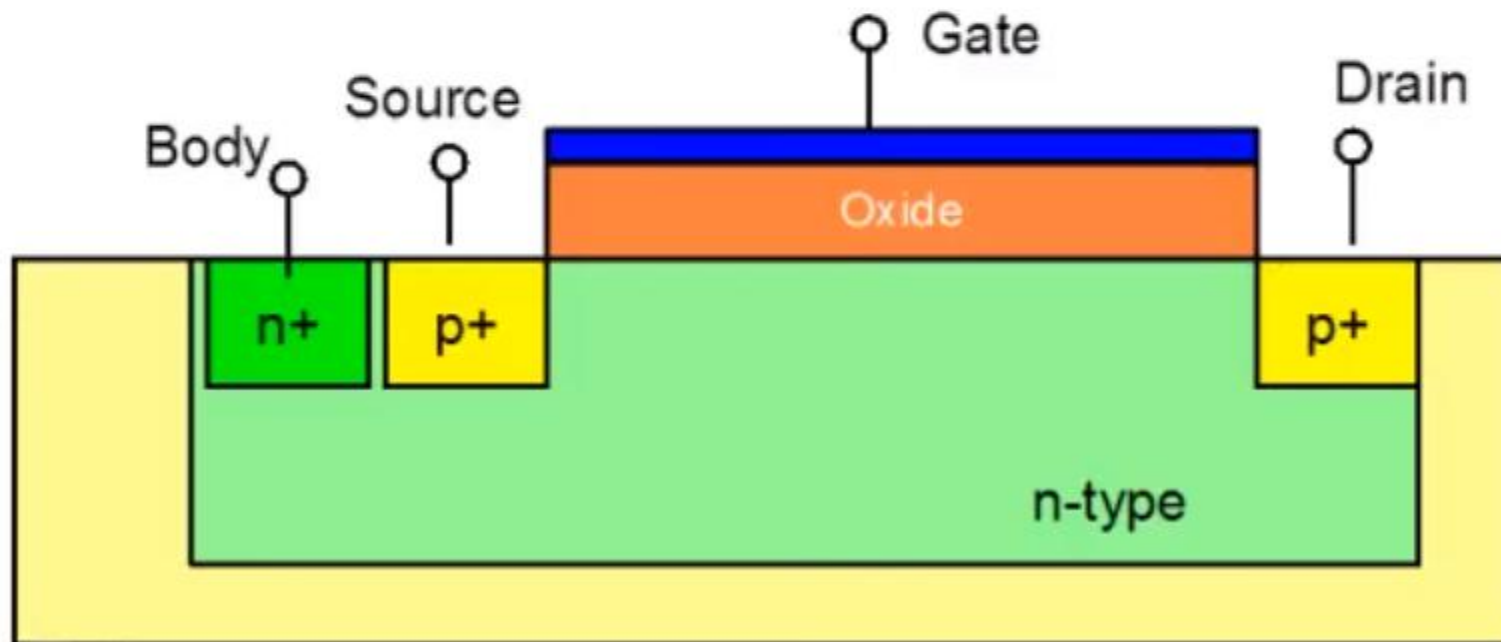
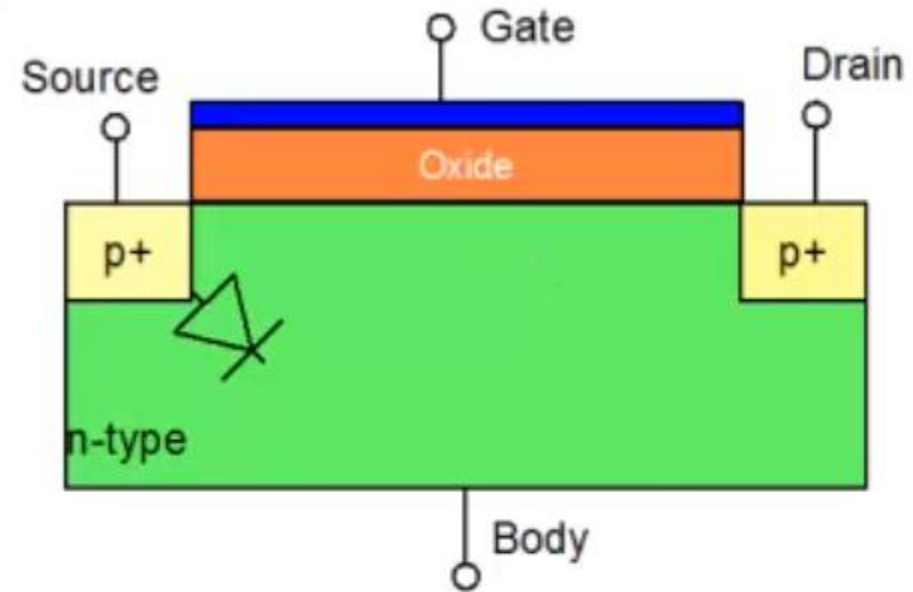
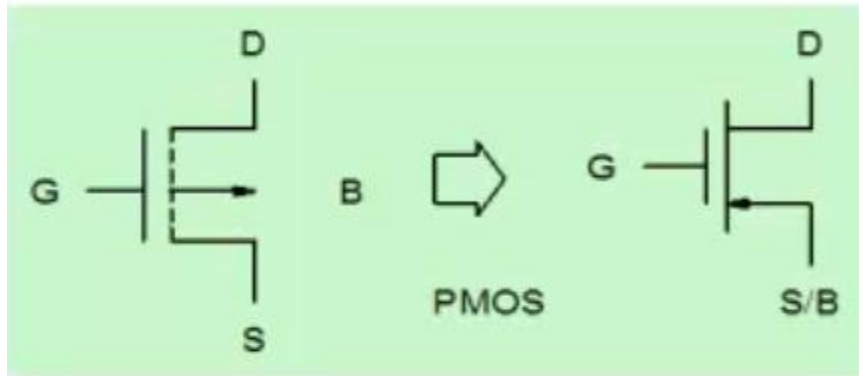
Metal Oxide Semiconductor Field Effect Transistor:

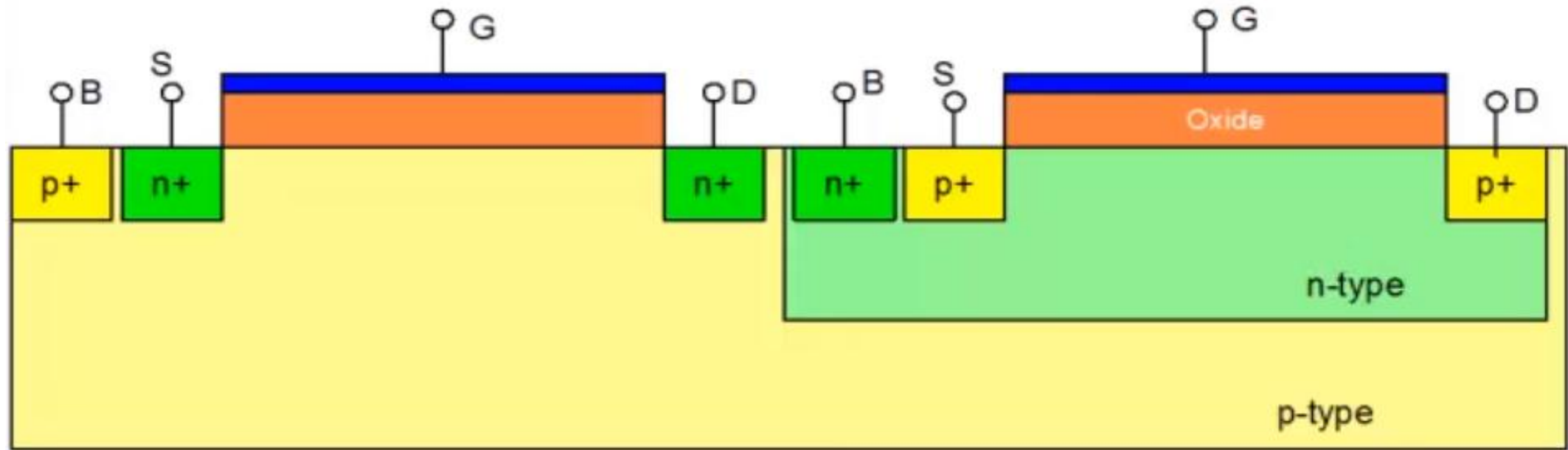


Simplified Symbols and structure

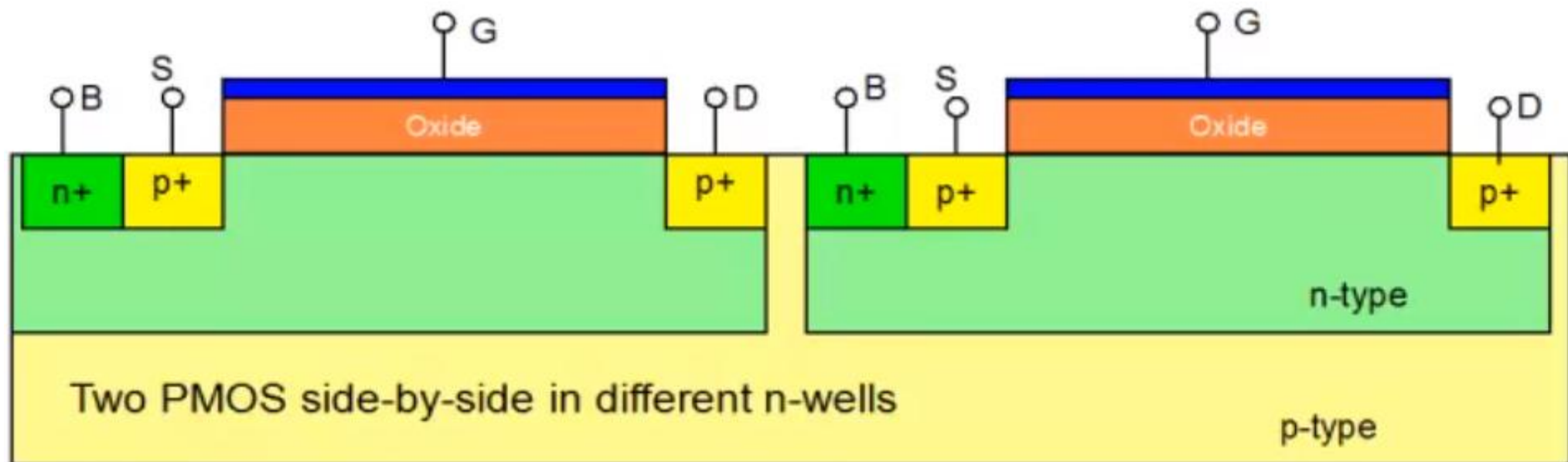
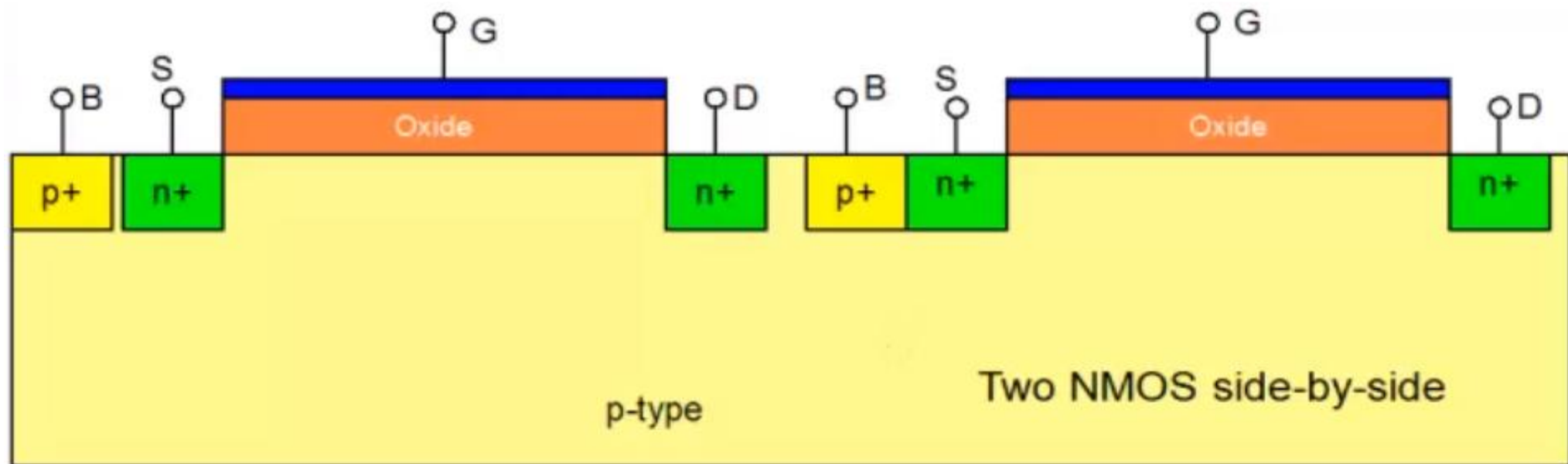


Simplified Symbols and structure

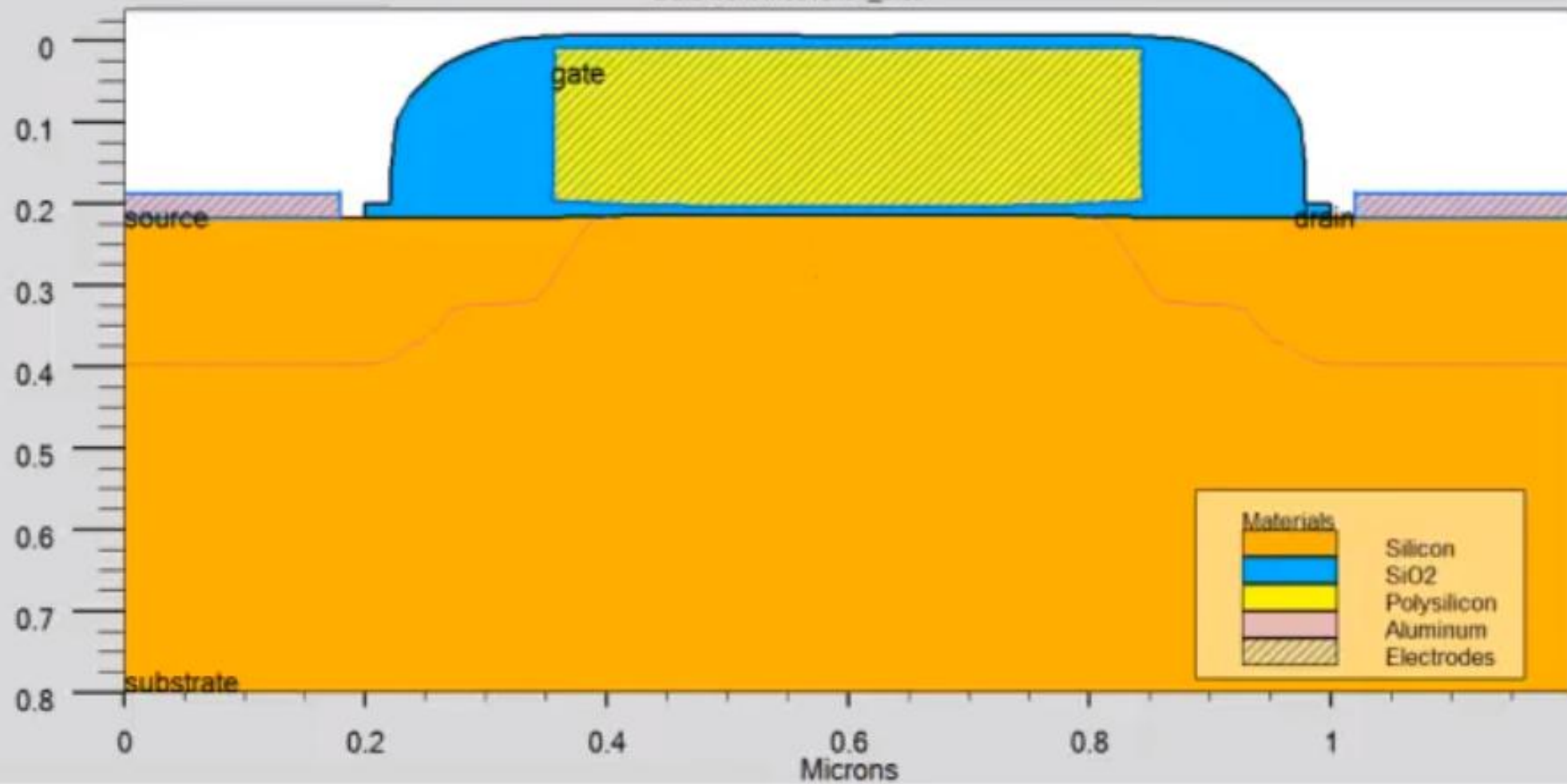


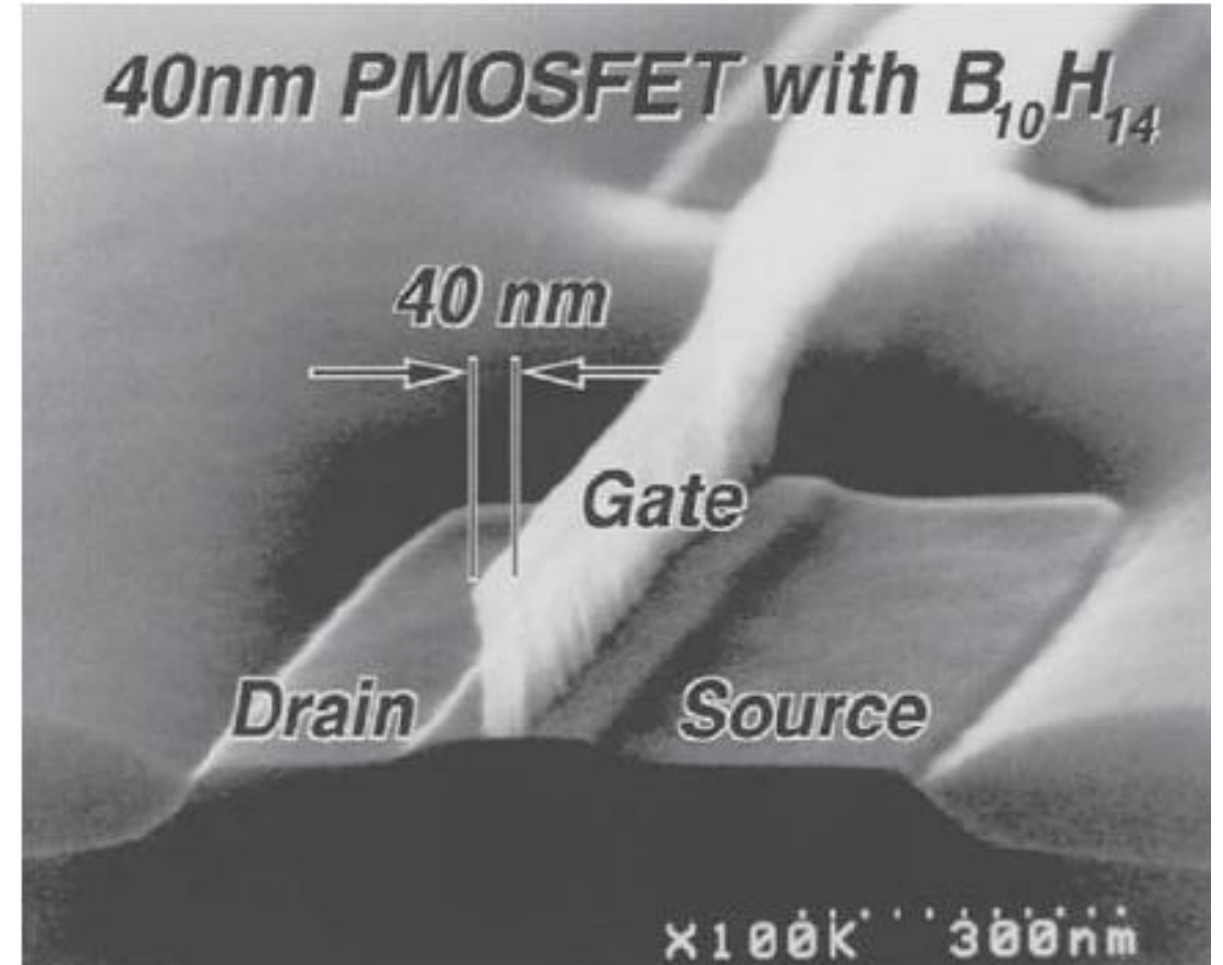
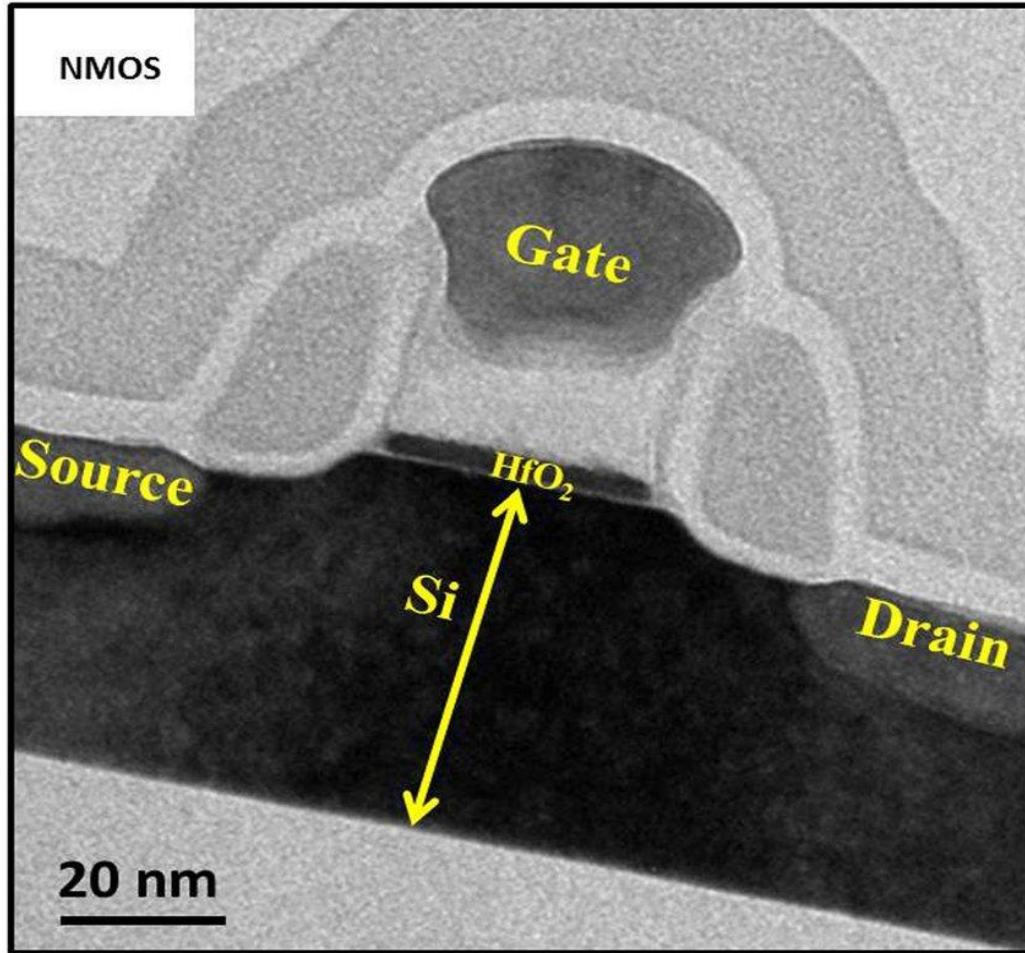


- Body potential of NMOS transistors is same and is normally connected to the most negative voltage in the circuit to ensure that $V_{BS} < 0$ and thus body-source PN junction is reverse biased.
- Each PMOS can be fabricated in a separate individual N-well and thus each pmos body terminal can have a distinct voltage. Normally body and source terminals of PMOS are shorted together.

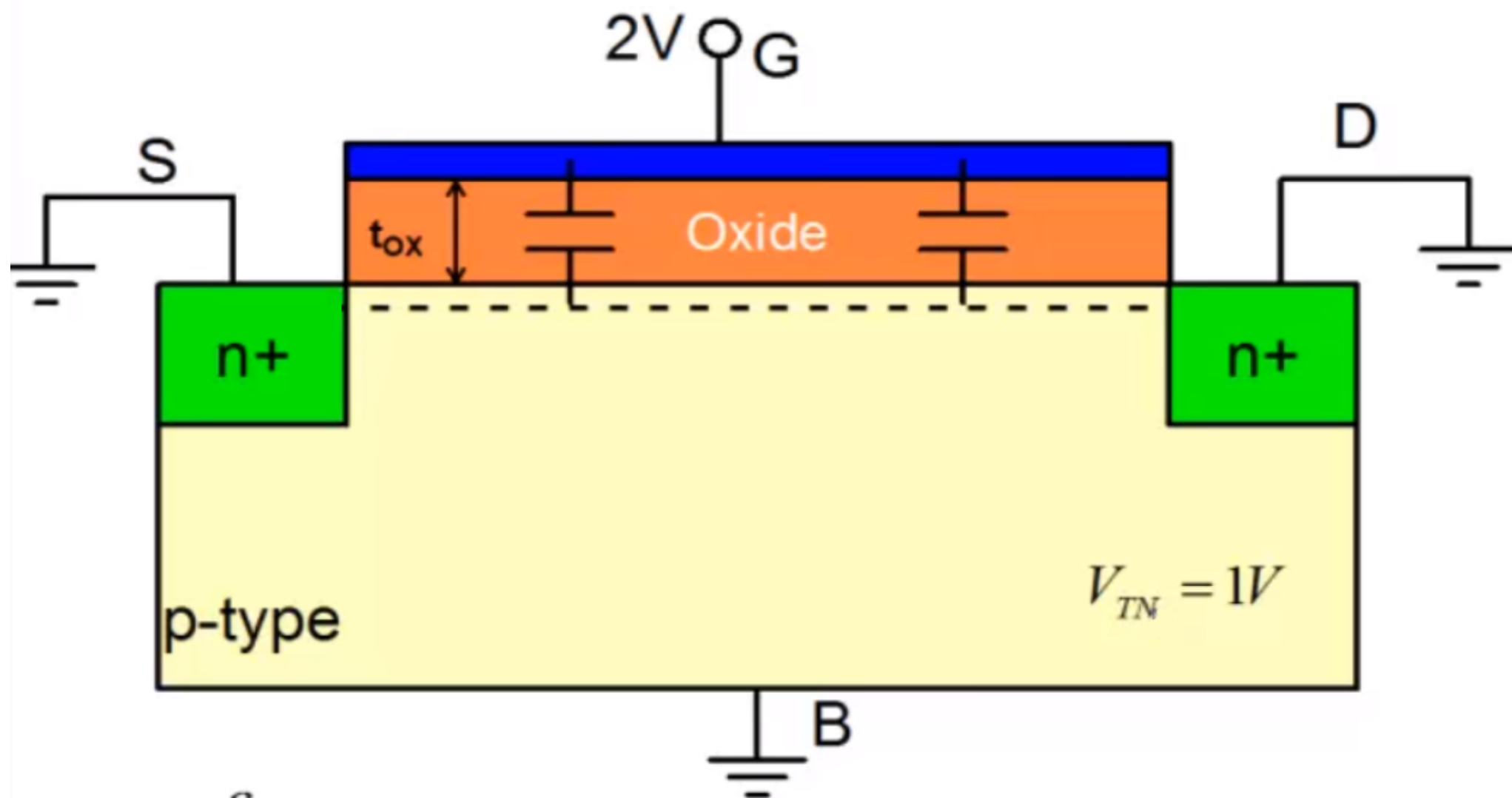


ATHENA
Data from mos1ex01_0.str



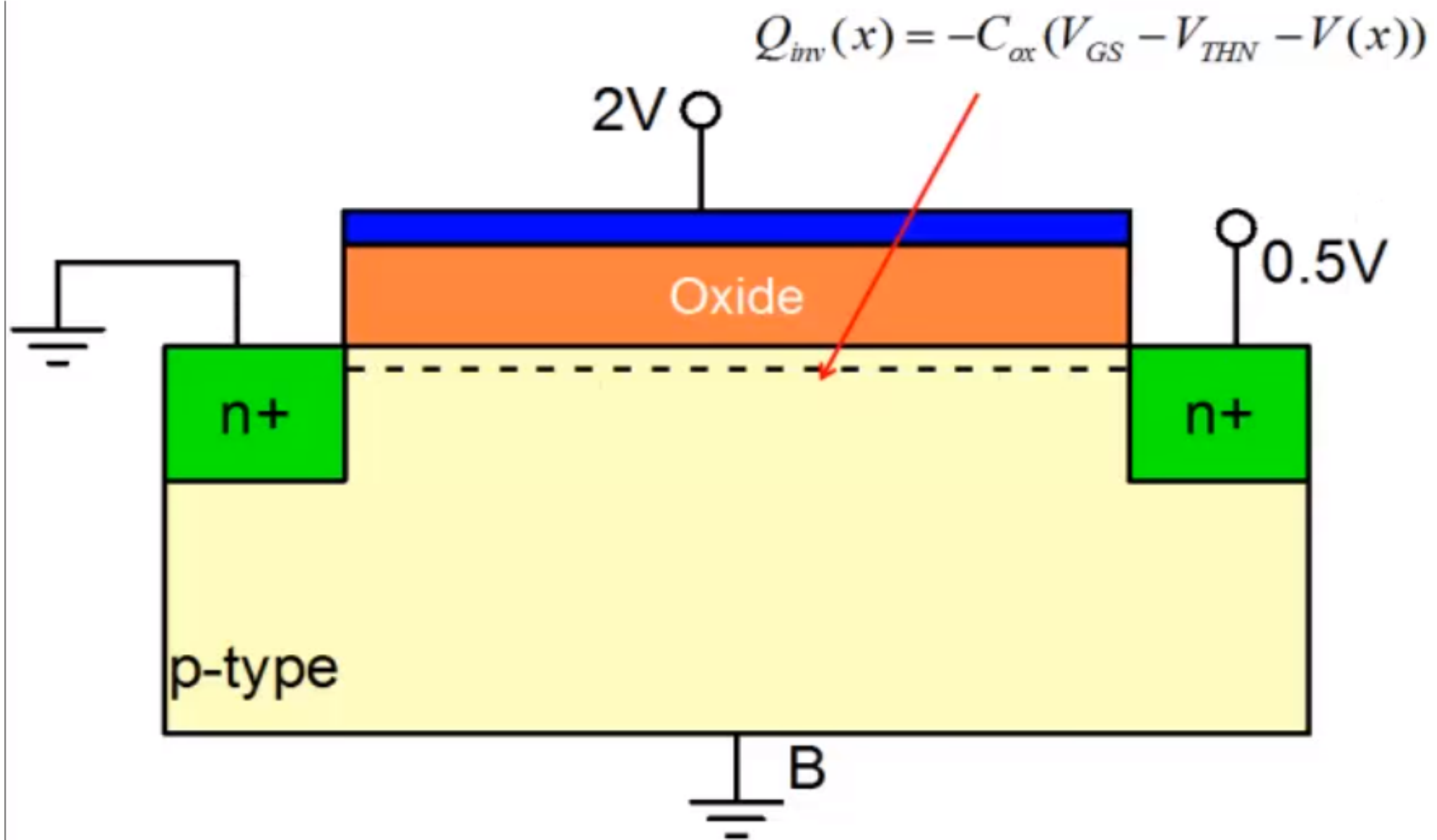


Operation of the MOSFET

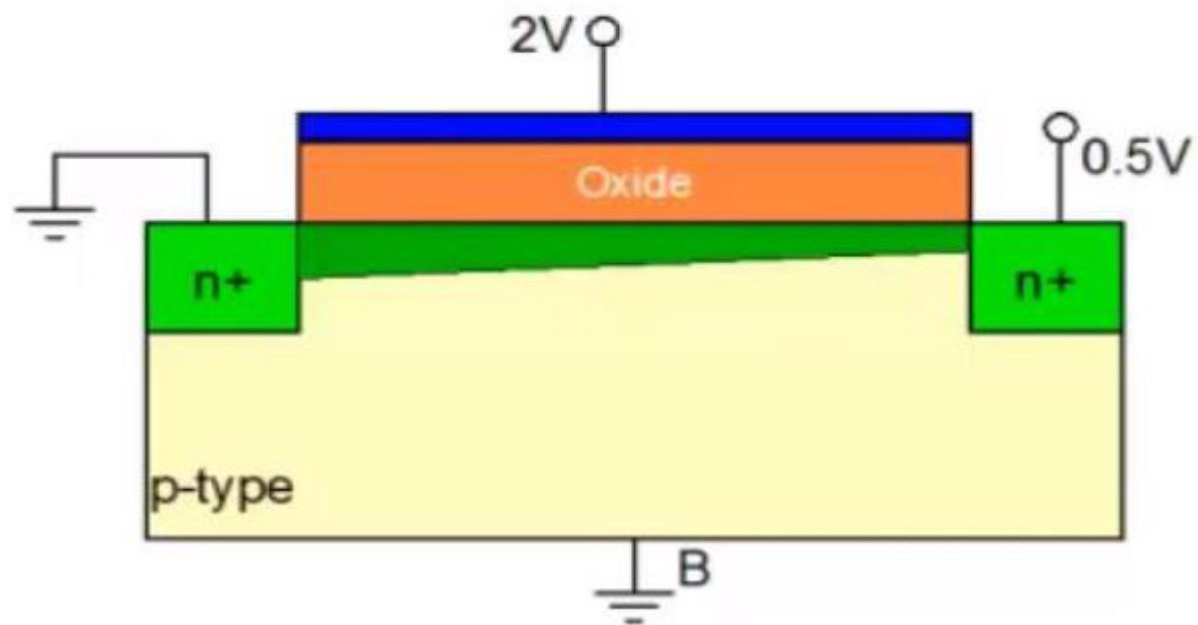
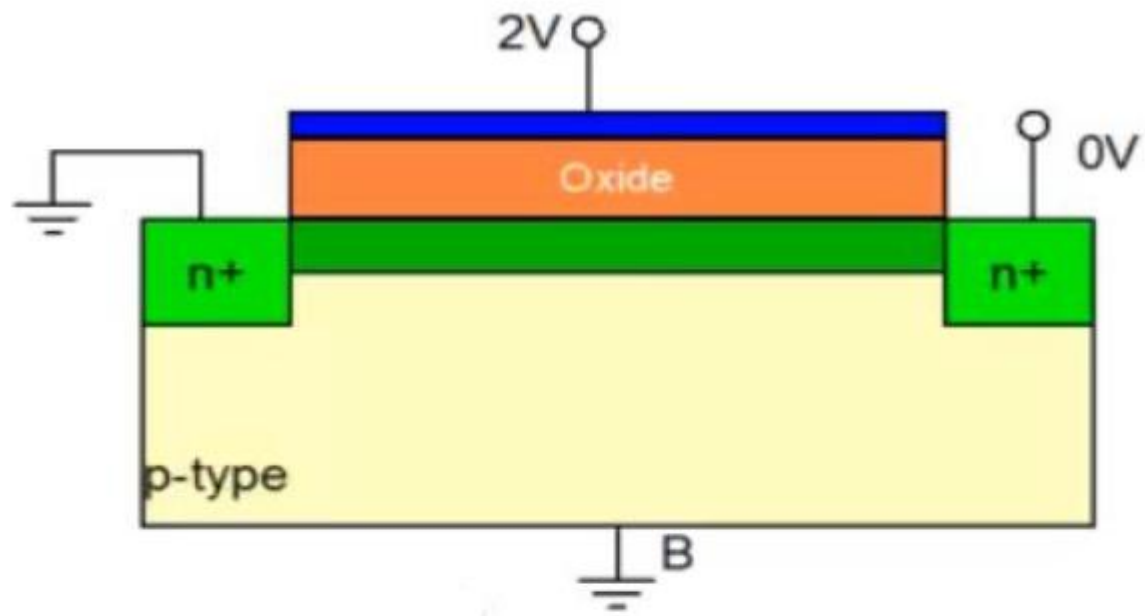


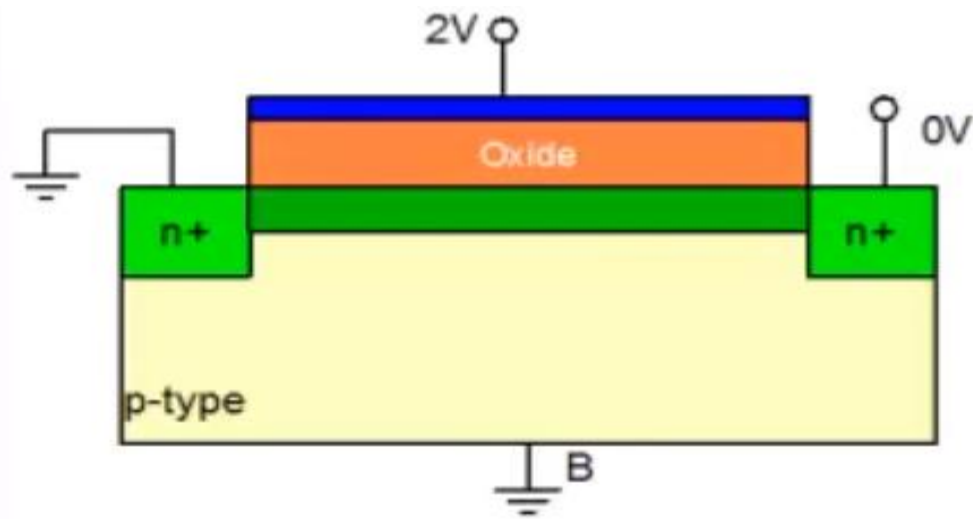
$$C'_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

Inversion charge/area : $Q_{inv} = -C_{ox}(V_{GS} - V_{THN})$



When a positive drain voltage is applied, current flows from drain to source and inversion charge density decreases from source to drain end.

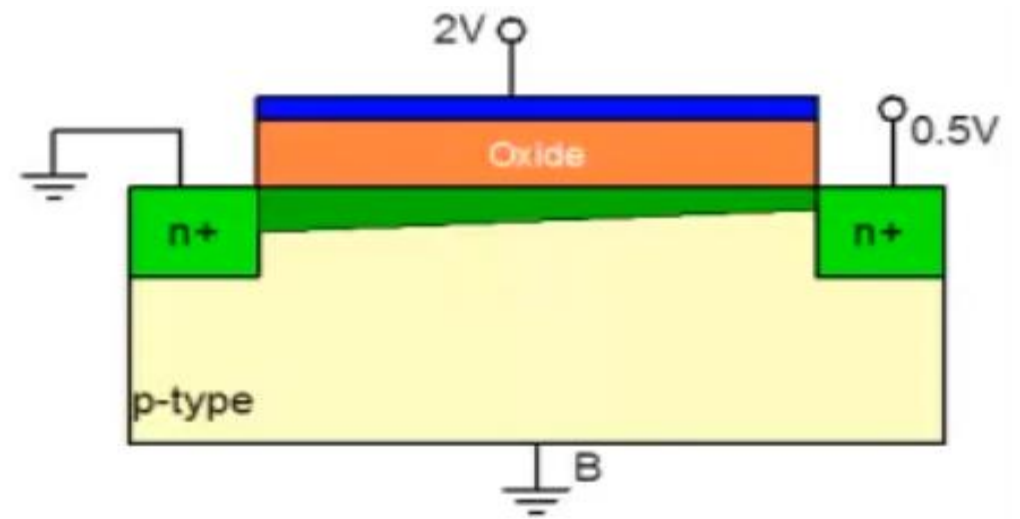




$$dR_{ch} = \frac{1}{q\mu n(x)} \frac{dx}{W \times t}$$

$$R_{ch} \propto \frac{1}{\int Q_{inv} dx}$$

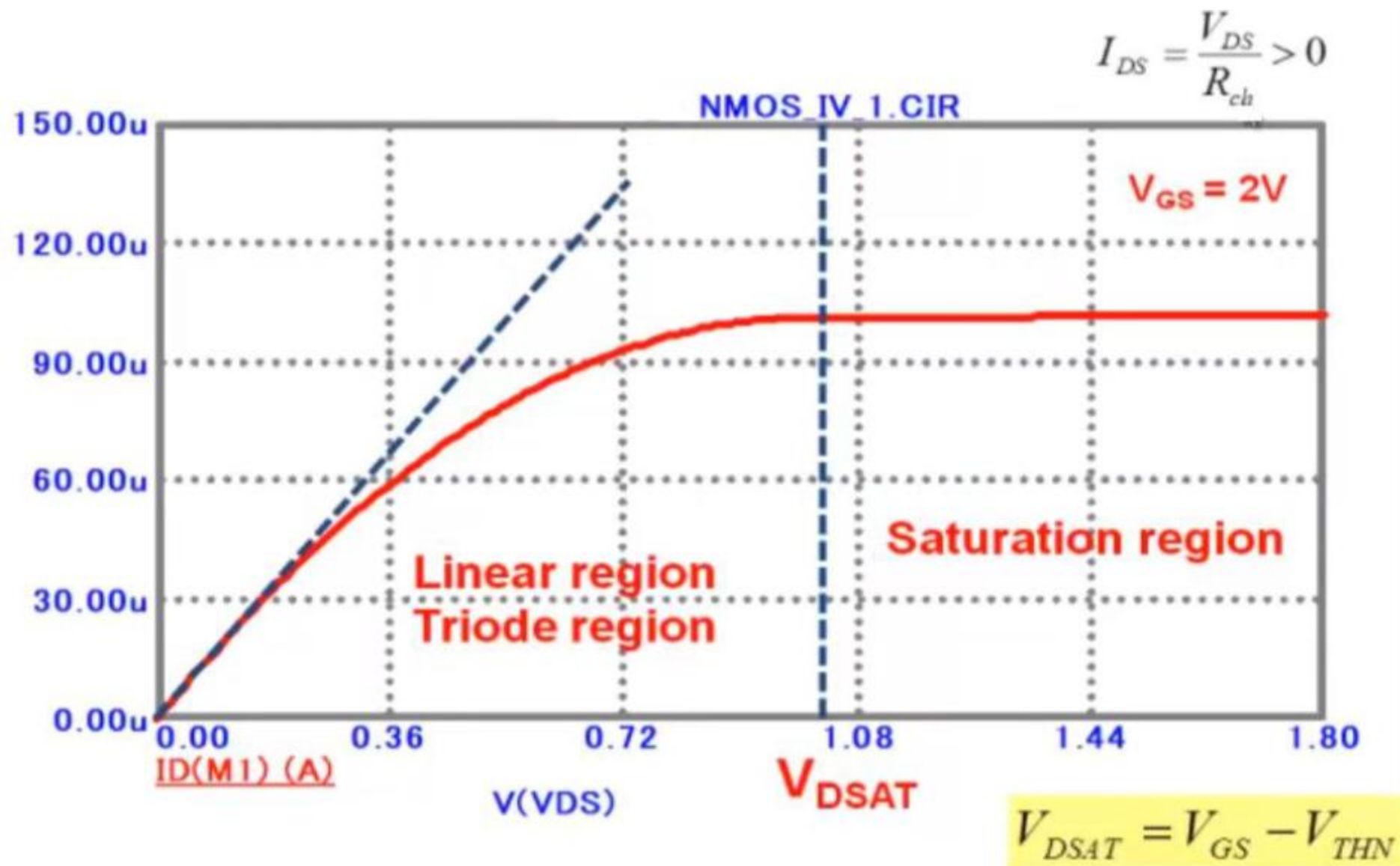
$$I_{DS} = \frac{V_{DS}}{R_{cho}} = 0$$



$$R_{ch} > R_{cho}$$

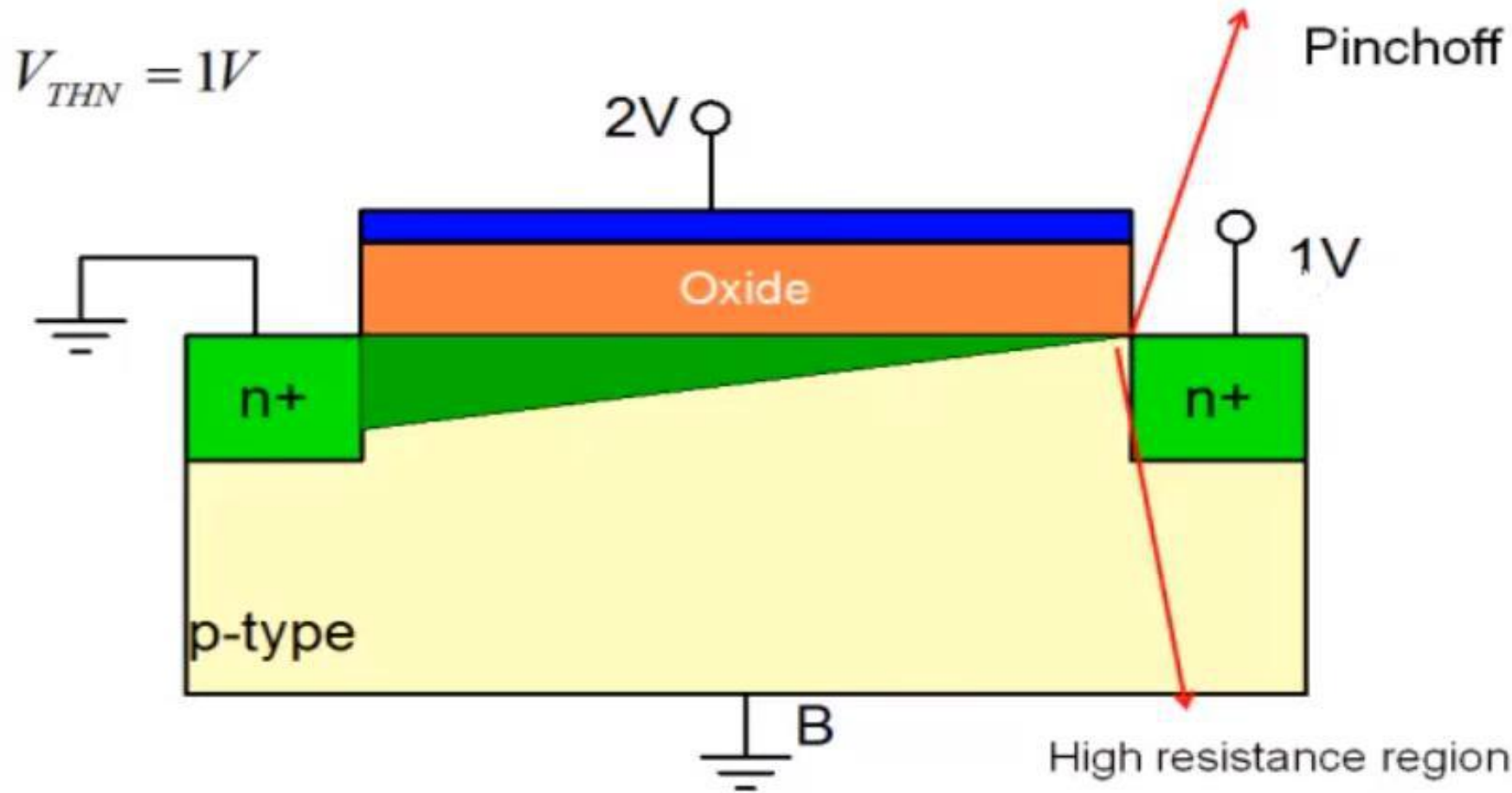
$$I_{DS} = \frac{V_{DS}}{R_{ch}} > 0$$

As drain voltage increases, channel resistance also increases causing drain current to depart from linear behavior

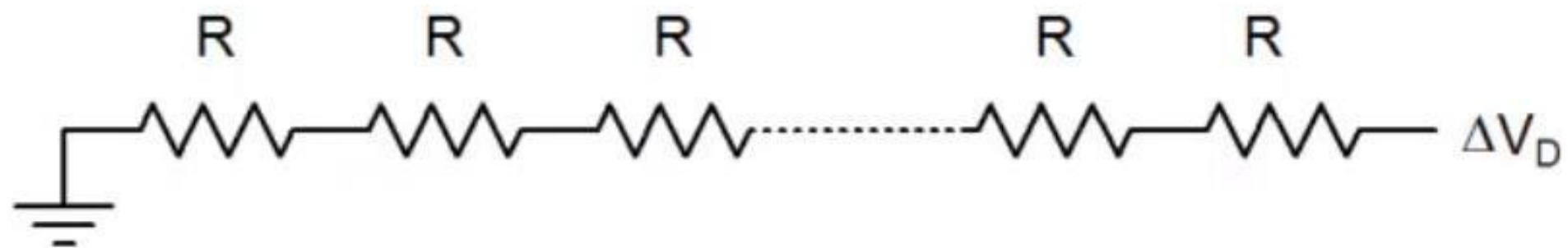


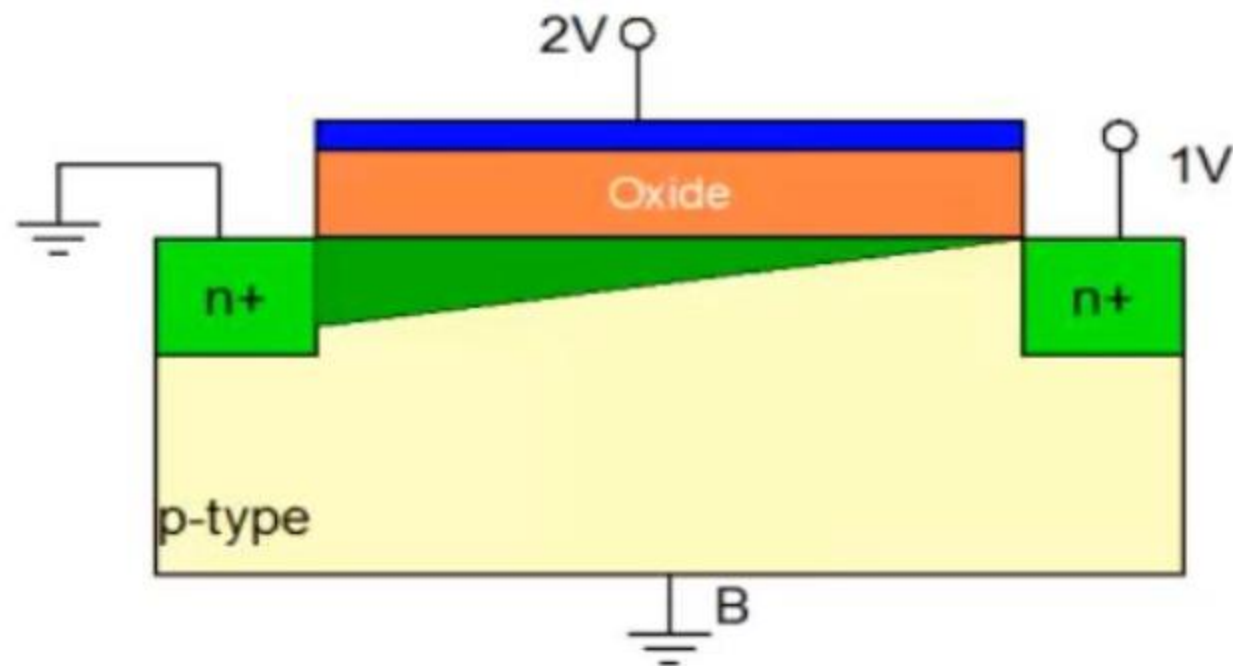
Note that **Saturation in MOSFET** is analogous to **forward active mode in BJT** and linear region is analogous to saturation in BJT.

$$Q_{inv}(x) = -C_{ox}(V_{GS} - V_{THN} - V(x)) \cong -C_{ox}(2 - 1 - 1) = 0$$

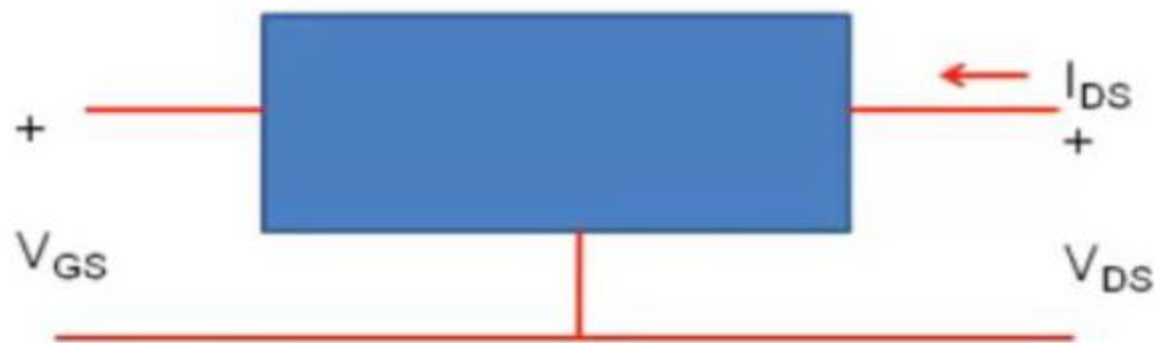


Any further increase in drain bias is absorbed in a small region next to the drain and rest of channel is not much affected and thus current becomes constant.





After pinchoff or saturation, drain current does not change much with drain voltage but is still very sensitive to gate voltage. MOSFET can now **AMPLIFY** signals



$$\frac{\partial I_{DS}}{\partial V_{GS}} \gg \frac{\partial I_{DS}}{\partial V_{DS}}$$